

ZYPHER

ZPH

A Healthcare Payment Protocol on Solana
Dual-Token Architecture for Equitable Provider Compensation

WHITEPAPER — DRAFT v0.4

June 2026

Blockchain: Solana Mainnet | Token Standard: SPL | Decimals: 9

Total Supply: 10,000,000,000 ZPH | TGE: March 17, 2026

Mint: 8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t

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1. Executive Summary

Zypher is a dual-token healthcare payment protocol built on Solana, designed to bridge the growing gap between the cost of clinical services and the ability of underinsured and uninsured patients to compensate their providers. The protocol's governance and utility token, ZPH, is live on Solana mainnet as of March 17, 2026. A companion stable settlement token, zCARE, will be developed in Phase 2 as a custom Anchor/Rust program to handle clinical payments at a stable value.

The United States healthcare system loses an estimated \$60–\$100 billion annually to uncompensated care — services rendered to patients who cannot pay, which providers are legally and ethically obligated to deliver. This burden falls disproportionately on independent practices, safety-net hospitals, and Federally Qualified Health Centers (FQHCs). Concurrently, the data infrastructure underlying provider networks — NPPES directories, CMS fee schedules, HRSA workforce files — is fragmented, inaccurate, and inaccessible in real time. Zypher addresses both problems simultaneously: broken payment rails and broken provider data infrastructure.

ZPH is currently live and tradeable on Raydium, discoverable via Jupiter aggregator, and tracked on DexScreener. The project website (zypherthecoin.github.io/Zypher) is live with a full How-to-Buy guide, tokenomics, and roadmap. CoinGecko and Jupiter VRFD listing submissions have both been completed. Community infrastructure — Twitter (@zypherthecoin), Telegram announcement channel, and community group — is active and growing. The immediate priority is growing total value locked (TVL) in the Raydium ZPH/SOL pool to the ~\$5,000 threshold required for CoinGecko listing activation.

2. The Problem: Healthcare's Payment Crisis

2.1 The Underinsured Gap

Approximately 27 million Americans lack health insurance entirely, while an estimated 40–45 million more are considered underinsured — meaning their coverage leaves them exposed to catastrophic out-of-pocket costs. For these patients, seeking care often means choosing between their health and financial ruin.

[TO WRITE: Statistics on medical debt, bankruptcy rates attributable to healthcare costs, and growth trends in high-deductible health plans. Source from KFF, Census Bureau, and CMS data.]

2.2 Provider Compensation Erosion

On the provider side, the economics are equally challenging. Uncompensated care costs U.S. hospitals and independent providers billions annually. Independent practices and smaller provider groups bear a disproportionate share of this burden, often absorbing the cost of treating patients who cannot pay or offering steep self-pay discounts that barely cover overhead.

[TO WRITE: Data on uncompensated care, charity care write-offs, provider burnout linked to financial stress, and the impact on rural/safety-net providers. Reference AHA, MGMA, and CMS Cost Reports.]

2.3 Broken Provider Data Infrastructure

The National Plan and Provider Enumeration System (NPPES) — the federal database of every licensed provider in the U.S. — is estimated to contain significant inaccuracies, with provider directory error rates often exceeding 50%. CMS fee schedules, HRSA workforce files, and state licensing board rosters are maintained in separate silos with no unified real-time query layer. Zypher aggregates and normalizes these data sources into a live, ZPH-gated provider intelligence layer.

2.4 Why Existing Solutions Fall Short

Current alternatives — sliding-scale fees, payment plans, medical credit cards, and health sharing ministries — each have significant limitations. Sliding-scale models reduce revenue. Payment plans carry high default rates. Medical credit cards impose predatory interest rates on vulnerable patients. None of these approaches fundamentally realign the incentive structure between patients and providers.

[TO WRITE: Detailed comparison table of current payment alternatives with failure modes for each.]

3. The Zypher Protocol: A Dual-Token Architecture on Solana

Zypher introduces two complementary tokens: zCARE (stable settlement token) and ZPH (ecosystem utility and governance token). Both are built on Solana's high-performance blockchain using the SPL token standard. This separation is deliberate — it isolates the payment stability providers need from the ecosystem dynamics that drive network growth.

3.1 ZPH: The Ecosystem Utility & Governance Token

ZPH is the native governance and utility token of the Zypher ecosystem. Deployed on Solana mainnet on March 17, 2026 via Smithii.io, with a total fixed supply of 10,000,000,000 tokens and 9 decimal places. ZPH is live and tradeable on Raydium and discoverable through Jupiter. Mint Authority, Freeze Authority, and Update Authority have all been permanently revoked — supply is fixed and metadata is immutable.

3.1.1 Deployment Specifications

Parameter	Value
Token Name	Zypher
Ticker Symbol	ZPH
Blockchain	Solana Mainnet
Token Standard	SPL Token
Total Supply	10,000,000,000
Circulating Supply	~1,735,000,000 (as of TGE)
Decimals	9
Token Generation Event	March 17, 2026
Mint Address	8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t
Deployment Platform	Smithii.io

Parameter	Value
Metadata Hosting	IPFS via Pinata (CID: bafkreihn5krmef7sbnsvnav7clvp6on3h5edarl7nudbvvyjlgxz4zx6q)
Initial Liquidity	Raydium ZPH/SOL pool
Raydium Pool ID	FKsYM7mBRR3oFPnJZYNYzSu9MjJLC4kYxatzYtPq8Aba
Aggregator Listing	Jupiter (jup.ag)
DEX Tracker	DexScreener
Project Website	zypherthecoin.github.io/Zypher
GitHub Repository	github.com/zypherthecoin/Zypher
Twitter / X	@zypherthecoin
Telegram Announcements	t.me/ZypherZPH
Telegram Community	t.me/ZypherCommunity
Mint Authority	Revoked — no new tokens can be created
Freeze Authority	Revoked — accounts cannot be frozen
Update Authority	Revoked — metadata permanently immutable

3.1.2 Utility Functions

ZPH is engineered as a multi-purpose token whose value derives from active use within the protocol rather than speculation alone. Its utility spans five categories.

Governance. ZPH holders direct the evolution of the protocol through on-chain voting. Eligible proposals include adjustments to fee parameters, the schedule and composition of zCARE reserves, the categories of providers eligible to participate, treasury disbursements, and the approval of new protocol features. Voting weight is proportional to staked ZPH, with safeguards (described in §11.3) to prevent capture during the progressive decentralization period.

Staking. Liquidity providers and network participants stake ZPH to support protocol operations and earn yield drawn from a combination of protocol fee revenue and ecosystem-development allocations. Staking also serves a stabilizing function: locked supply reduces circulating ZPH, and longer lock durations earn proportionally higher rewards.

Data access. ZPH functions as the access key to the Zypher Intelligence Layer (§7). Holding or staking ZPH unlocks tiered access to NPI-linked provider profiles, credentialing and licensure status, network-adequacy analytics, and workforce data built on NPPES, CMS, and HRSA foundations. Higher tiers expose richer query volumes and advanced analytics.

Provider credentialing. ZPH is the unit of payment for on-chain credential-verification services, allowing providers to establish and maintain a verifiable, portable credential record within the network.

Fee discounts. Holding ZPH reduces transaction and redemption fees across the network, aligning the incentives of active participants with long-term token retention.

3.1.3 Supply and Distribution

ZPH has a fixed total supply of 10,000,000,000 tokens with mint authority permanently revoked. The table below reflects the intended long-term allocation plan. Current on-chain distribution is provided separately in Section 3.1.4.

Category	Intended Allocation	% Total	Cliff	Vesting
Team & Founders	1,500,000,000	15.0%	12 mo	48 mo
Advisors	500,000,000	5.0%	6 mo	24 mo
Ecosystem Development	2,000,000,000	20.0%	None	60 mo
Community & Rewards	1,500,000,000	15.0%	None	60 mo
Treasury / DAO	2,000,000,000	20.0%	None	Gov-controlled
Liquidity Provision	1,000,000,000	10.0%	None	12 mo
Public Sale (TGE)	1,000,000,000	10.0%	None	Immediate
Private/Seed Investors	500,000,000	5.0%	6 mo	36 mo

3.1.4 Current On-Chain Distribution

The following reflects actual on-chain holder distribution as verified on Solscan at the time of CoinGecko listing submission. All allocations are currently held in the project treasury wallet pending governance-controlled distribution per the plan above.

Holder / Wallet	ZPH Balance (approx.)	% of Supply	Classification
Project Treasury / Reserve Wallet	~8,265,000,000	82.65%	Locked — Project Treasury
Raydium ZPH/SOL Liquidity Pool	~1,463,000,000	14.63%	Active — LP Deployed
Public Market Holders	~272,000,000	~2.72%	Circulating
TOTAL CIRCULATING SUPPLY	~1,735,000,000	17.35%	Non-treasury

i Transparent disclosure: The majority of supply is held in the project treasury wallet with no on-chain vesting enforcement. This reflects the early-stage nature of the protocol. As governance infrastructure matures, distributions will be executed per the allocation plan above. All supply and holder data can be independently verified via Solscan.

3.1.5 Value Accrual

ZPH is designed to accrue value through four reinforcing mechanisms rather than through artificial scarcity alone.

Fee capture and buyback-and-burn. A defined percentage of all zCARE transaction fees is routed to the protocol treasury and used to buy ZPH on the open market (via the Raydium

ZPH/SOL pool) and permanently burn it. As clinical settlement volume grows, this creates persistent buy-side pressure and a deflationary supply effect tied directly to real protocol usage.

Data demand. As the provider intelligence layer matures and its datasets deepen, demand for ZPH-gated data access rises. Because access is metered in ZPH, growth in the analytics business translates into growth in token demand.

Network-effects flywheel. Each additional provider increases the value of the network to patients, and each additional patient increases the value to providers. More participants generate more transactions (fee capture), more credentialing activity, and more data demand — each of which strengthens the others, producing a compounding flywheel rather than linear growth.

Staking lockup. Staking incentives remove ZPH from circulating supply for extended periods. By reducing float while utility-driven demand grows, staking amplifies the value impact of the buyback-and-burn and data-demand mechanisms.

3.2 zCARE: The Stable Settlement Token (Phase 2)

zCARE is the unit of account for clinical transactions within the Zypher network. Unlike ZPH, which is deployed as a standard SPL token, zCARE requires custom Solana program development (Rust/Anchor framework) to implement mint/burn authority, oracle integration, and reserve management. zCARE development will commence once ZPH governance is operational and initial protocol revenue supports the engineering investment.

3.2.1 Peg Mechanism

zCARE is the protocol's stable unit of account for clinical settlement, and its peg design is the single most consequential engineering decision in Phase 2. Three approaches are under consideration.

Option A — Hard USD peg (1 zCARE = \$1.00). zCARE is fully backed by reserves held in USD-equivalent assets (USDC on Solana, T-bill-backed stablecoins, and a modest overcollateralized crypto buffer). This is the simplest model to communicate to providers and patients, who price care in dollars, and it is the most defensible from a redemption standpoint. The tradeoff is reserve cost and the operational burden of maintaining full backing and attestations.

Option B — CMS Fee Schedule Index peg. zCARE is pegged to a weighted basket of Medicare reimbursement rates rather than the dollar, so that one zCARE tracks a defined unit of clinical value over time. This naturally aligns the token with healthcare cost inflation (\$4.4) and insulates providers from the erosion of fixed-dollar settlement. The tradeoff is complexity: it requires a reliable CMS-data oracle, a transparent index methodology, and clear communication, and it introduces basis risk against the dollar that liquidity providers must hedge.

Option C — Hybrid. zCARE maintains a hard USD peg for day-to-day settlement and redemption, while a separate index reference (the CMS basket) governs periodic, governance-approved adjustments to provider-facing pricing. This captures most of the inflation-alignment benefit while preserving the simplicity and redeemability of a dollar peg.

Recommended direction. A hard USD peg at launch (Option A) minimizes adoption friction and regulatory ambiguity, with a defined governance pathway toward the hybrid model (Option C) once volume and reserves justify the added complexity.

Reserve and oracle requirements. Regardless of option, reserves should target $\geq 100\%$ backing for a hard peg (with an overcollateralization buffer for any volatile component), undergo regular third-party attestation, and be priced through redundant Solana-native oracles — Pyth Network as primary with Switchboard as fallback — using a defined update cadence and staleness checks to prevent settlement against stale or manipulated feeds.

3.2.2 Minting and Redemption

zCARE is created and destroyed through a custom Solana program whose mint and burn authority is held by the protocol multisig rather than any single party.

Minting. Authorized on-ramp partners and liquidity providers deposit eligible collateral (primarily USDC on Solana) into the reserve, and the program mints zCARE 1:1 against that deposit, less the minting fee. Mint authority is constrained on-chain so that zCARE cannot be issued in excess of verified reserves.

Collateral. The reserve is held in the composition described in §4.2 and is continuously verifiable on-chain and through periodic attestation. Collateralization ratios and eligible asset types are governance parameters adjustable by ZPH holders.

Redemption. Providers holding zCARE redeem it for USDC (and onward to USD via Solana-native stablecoin rails) through the same program, which burns the redeemed zCARE and releases the corresponding reserves, less the redemption fee. Redemptions are processed within defined settlement windows to manage reserve liquidity, with an expedited path available to providers meeting volume thresholds.

Authorized liquidity providers (ALPs). A vetted set of ALPs guarantees two-sided liquidity, ensuring providers can always redeem at par. ALPs earn a share of mint and redemption fees in exchange for maintaining inventory and absorbing short-term settlement timing mismatches.

3.2.3 Transaction Flow

The end-to-end settlement path is designed to feel, from the provider's perspective, no more complex than accepting a card payment.

- **Acquisition:** A patient acquires zCARE through one of the channels in §5 (direct purchase, employer funding, grant distribution, or earned models) and holds it in a Solana wallet.
- **Point of service:** At the point of care, the patient presents zCARE — via wallet, QR code, or an integrated patient-facing app — and the agreed amount is transferred to the provider's wallet on-chain, with sub-second finality.
- **Record:** The provider's practice-management or billing system records the zCARE payment against the encounter, associating it with the relevant CPT/HCPCS codes and reflecting it on the patient's statement and EOB workflow.
- **Disposition:** The provider either holds zCARE for ecosystem use or redeems it for USDC/USD (§3.2.2) on their preferred settlement cadence.

Billing integration. The flow is designed to map onto existing revenue-cycle workflows: zCARE payments carry metadata linking them to procedure codes, support reconciliation against claims, and produce the documentation providers already generate for self-pay and

patient-responsibility balances. A reference integration diagram and PMS/EHR connector specification will accompany the Phase 2 technical documentation.

4. Token Economics Deep Dive

4.1 Economic Model

Attribute	zCARE (Settlement)	ZPH (Utility/Gov)
Primary Use	Clinical payments	Governance, staking, data access
Blockchain	Solana (custom program)	Solana (SPL token)
Value Target	Stable (\$1 USD or CMS-indexed)	Market-driven
Total Supply	Elastic (mint/burn on demand)	10,000,000,000 (fixed cap)
Decimals	6 (USDC-compatible)	9 (Solana standard)
Who Holds It	Patients, providers, payers	Ecosystem participants, investors
Volatility	Near-zero (by design)	Variable (market forces)
Revenue Link	Transaction volume	Fee buyback/burn + data demand
Trading Venue	Protocol-only (not speculative)	Raydium, Jupiter
Development Status	Phase 2 — in planning	Phase 1 — live on Solana mainnet
TGE Date	TBD (post Phase 2 dev)	March 17, 2026

4.2 Stability Mechanism Design (zCARE)

zCARE's stability rests on full, transparent backing and redundant safeguards — deliberately avoiding the algorithmic, undercollateralized designs that have failed elsewhere.

(a) Reserve composition. Reserves are held predominantly in USDC on Solana and short-duration T-bill-backed stablecoins, with a limited, overcollateralized SOL buffer to provide on-chain liquidity for rapid redemptions. Target backing is $\geq 100\%$ of outstanding zCARE, with the buffer sized to absorb the volatility of any non-dollar component.

(b) Oracle design. Price feeds are sourced from Pyth Network as primary and Switchboard as fallback, with cross-checks, staleness thresholds, and deviation bounds. No mint, redemption, or reserve revaluation executes against a single unconfirmed feed.

(c) CMS data oracles. For any index-linked functionality, a custom oracle publishes CMS Fee Schedule data on-chain through a verifiable, governance-auditable pipeline, with a defined update cadence and provenance record.

(d) Arbitrage incentives. Authorized liquidity providers profit by minting when zCARE trades above par and redeeming when it trades below, continuously pulling the market price back to peg. Fee structures are tuned so that this arbitrage remains profitable at small deviations.

(e) Emergency mechanisms. The system includes circuit breakers that pause minting or redemption on abnormal feed deviation, a redemption queue to manage reserve liquidity under stress, and multisig governance intervention as a last resort, all with on-chain transparency.

(f) Lessons from prior failures. Unlike UST/Luna — an undercollateralized, reflexive algorithmic design — zCARE is fully reserve-backed and redeemable, with no dependence on the value of ZPH to maintain its peg. The separation of zCARE (backed settlement token) from ZPH (ecosystem token) is precisely what prevents the death-spiral dynamic that collapsed UST. zCARE's model is closer in spirit to fully-reserved stablecoins like USDC and PYUSD than to algorithmic designs.

4.3 Fee Structure

Fees are structured so that the all-in, round-trip cost of a zCARE clinical payment is competitive with card processing while remaining sustainable for the protocol.

The model comprises a minting fee (charged on entry of collateral), a transaction fee (charged on point-of-service settlement), and a redemption fee (charged when providers convert zCARE to USDC/USD). The target combined round-trip cost is 2.5–3.5%, in line with typical merchant card-processing costs, but with the crucial difference that Solana's base transaction cost (~\$0.00025) makes the on-chain component effectively negligible — so nearly all of the fee funds reserves, liquidity provision, and the buyback-and-burn rather than network rent.

Because the marginal on-chain cost is near zero, the protocol can support microfee structures and volume-tiered pricing: high-volume providers and ZPH holders receive reduced rates, while baseline rates apply to occasional users. A full fee-scenario model — showing effective round-trip cost at \$20, \$50, \$100, and \$200 payment sizes across low, medium, and high monthly volumes — will accompany the tokenomics workbook (Appendix B).

4.4 Inflation, Deflation, and Healthcare Cost Alignment

Healthcare costs inflate at approximately 3–5% annually. ZPH has a fixed 10B supply cap with a 2.5% annual ecosystem fund emission rate, offset by a buyback-and-burn mechanism funded by 30% of protocol fee revenue. As transaction volume grows, burn pressure is designed to exceed inflation, creating net deflationary dynamics.

[TO WRITE: Model the inflation/burn interaction and ZPH circulating supply over 5 years. Sensitivity to ZPH price and transaction volume assumptions.]

5. Patient Token Acquisition Pathways

The fundamental challenge: if patients lack cash to pay providers directly, how do they acquire zCARE? Zypher addresses this through multiple acquisition channels. Note: patient acquisition of zCARE is a Phase 2 capability. In Phase 1, ZPH serves as the ecosystem bootstrapping mechanism and can be acquired today via Raydium or Jupiter using SOL.

5.1 Direct Purchase

In Phase 1, ZPH is accessible via Raydium (direct) or Jupiter aggregator. Users can acquire SOL via Coinbase, Kraken, or Cash App, then swap to ZPH. A full 6-step How-to-Buy guide

(from wallet setup through Jupiter swap) is published on the project website. Phase 2 will introduce dedicated fiat-to-zCARE rails via MoonPay/Transak integration.

5.2 Employer-Sponsored Programs

[TO WRITE: Employers fund zCARE wallets as a supplemental health benefit. Payroll integration. Section 125 cafeteria plan eligibility, HSA coordination.]

5.3 Community Health Grants & Government Programs

[TO WRITE: FQHC and safety-net provider partnerships. Grant-funded zCARE distribution. HRSA-funded program integration. State Medicaid waiver pilot opportunities.]

5.4 Earned Token Models

[TO WRITE: Patients earn zCARE through verified health-promoting behaviors: preventive care, chronic disease management compliance, health literacy completion. Verification through provider attestation or connected health device APIs.]

5.5 Charitable and Philanthropic Channels

[TO WRITE: Nonprofits, faith-based organizations, and foundations purchase zCARE in bulk for distribution. Donor tax treatment.]

6. Provider Compensation & Redemption

6.1 Provider Onboarding

[TO WRITE: NPI-verified provider enrollment. Phantom wallet setup or custodial option. Provider types eligible. Credentialing via NPPES, state licensing boards, and DEA databases.]

6.2 Point-of-Service Workflow

[TO WRITE: How a zCARE payment works at the front desk. ~400ms Solana confirmation. Integration options: standalone Zypher terminal/app, EHR/PMS plugin, clearinghouse integration.]

6.3 Redemption Options

[TO WRITE: Off-ramps: (a) USDC via Raydium/Jupiter → fiat via Coinbase/Kraken. (b) Hold zCARE. (c) Convert to ZPH for staking. (d) Staff bonuses in zCARE.]

6.4 Provider Incentive Structure

Provider participation is the linchpin of the network, and Zypher offers providers concrete, self-interested reasons to join rather than relying on goodwill.

Providers gain access to otherwise-unreachable patients — the underinsured and self-pay populations who would typically generate uncompensated care or bad debt — now arriving with a funded, redeemable payment instrument. Settlement is guaranteed and fast: zCARE redeems at par through the reserve, removing the default and collection risk inherent in payment plans and self-pay balances. Costs are lower than medical credit cards and competitive with ordinary card processing. Early-adopting providers earn ZPH staking rewards and preferential fee tiers. Participation also confers access to the provider intelligence and credentialing tools of §7 —

network-adequacy visibility, portable on-chain credentials, and analytics that have standalone value independent of payment volume. Taken together, these turn participation from a charitable gesture into a rational business decision.

7. The Zypher Intelligence Layer

A key differentiator for Zypher is the integration of a comprehensive provider intelligence platform powered by ZPH token access. This layer transforms Zypher from a simple payment rail into a healthcare data ecosystem, with federal datasets as its foundation.

7.1 Data Sources & Integration

- **NPPES:** NPPES (National Plan and Provider Enumeration System) — every licensed U.S. provider, updated weekly. Source of truth for NPI-linked provider identity.
- **CMS:** CMS Fee Schedules — Medicare/Medicaid reimbursement rates by CPT/HCPCS code, used to price-benchmark zCARE clinical transactions.
- **HRSA:** HRSA Area Health Resource Files — county-level workforce supply/demand, HPSA and MUA designations.
- **State Boards:** State licensing board rosters — supplemental credentialing data by specialty and jurisdiction.
- **CMS Utilization:** CMS Medicare utilization files — provider-level utilization patterns for network analytics.

7.2 Provider Profiles & Credentialing

[TO WRITE: On-chain provider identity tied to NPI. Credential verification stored as Solana account metadata or via Arweave/Shadow Drive. Real-time updates from licensing board integrations.]

7.3 Network Adequacy Analytics

[TO WRITE: Workforce supply/demand mapping by specialty and geography. HRSA HPSA/MUA overlay. ZPH-gated access for health plans and state regulators.]

7.4 Data Privacy & Compliance

[TO WRITE: HIPAA considerations. Provider data (public via NPPES) vs. patient data (PHI, never on-chain). Zero-knowledge proof options via Light Protocol on Solana.]

8. Technical Architecture

8.1 Why Solana

Zypher's requirements — point-of-care confirmation in seconds, per-transaction costs low enough to make \$20–\$200 payments rational, and throughput sufficient for national scale — narrow the viable platform set considerably.

Versus Ethereum L2s. Rollups inherit settlement dependence on Ethereum L1 and exhibit variable fees and finality, with fragmentation across competing L2s complicating a unified

payment network. Solana's monolithic high-throughput design delivers consistent sub-second finality (~400ms) and uniform low cost (~\$0.00025 per transaction) without cross-L2 liquidity fragmentation.

Versus Cosmos appchains. A sovereign appchain offers control but imposes the burden of bootstrapping and securing an independent validator set and liquidity, and complicates access to the deep stablecoin liquidity (USDC, PYUSD) that zCARE redemption depends on. Solana provides that liquidity and a mature SPL ecosystem out of the box.

Reliability concerns and mitigation. Solana has historically experienced network congestion and outage events. Zypher mitigates this through redemption queues and circuit breakers (§4.2) that fail safe during degraded conditions, redundant RPC and oracle infrastructure, retry-and-reconcile logic in the settlement layer, and — because zCARE is fully reserve-backed — no risk that a temporary halt impairs the peg. For healthcare-grade uptime, the protocol relies on geographically distributed, professionally operated validators and monitors network health as an operational SLA.

8.2 Smart Contract Architecture

The Zypher protocol is deployed across multiple Solana programs in phases.

Phase 1 (deployed). The ZPH SPL token, deployed via Smithii.io on March 17, 2026, with initial liquidity on Raydium (ZPH/SOL) and aggregator routing through Jupiter. All authorities permanently revoked.

Phase 2 (in development). (a) The zCARE token program — a custom Anchor/Rust program holding mint and burn authority under the protocol multisig. (b) The stability module, managing reserves and integrating Pyth oracle feeds with the circuit breakers and redemption queue of §4.2. (c) The provider registry, mapping verified NPI-linked identities to on-chain addresses and credential records. (d) The fee and treasury module, routing protocol fees to reserves and the buyback-and-burn mechanism. (e) Governance and staking programs for ZPH voting and reward distribution.

All custom programs will undergo independent third-party security audits and formal review of mint/burn authority constraints prior to mainnet deployment, with upgrade authority held by the multisig during the progressive-decentralization period and a defined path to immutability or DAO control.

8.3 Oracle Design

[TO WRITE: Pyth Network for ZPH/USD and SOL/USD feeds. Switchboard for custom feeds. Custom CMS data oracle for fee schedule updates.]

8.4 Interoperability

[TO WRITE: HL7 FHIR integration for EHR connectivity. X12 EDI compatibility. Wormhole bridge for cross-chain settlement. Solana Pay QR code integration.]

9. Regulatory & Compliance Framework

9.1 Securities Classification

[TO WRITE: Howey test analysis for zCARE and ZPH. ZPH as utility/governance token. Reference SEC guidance, GENIUS Act, STABLE Act, and state digital asset frameworks.]

9.2 Money Transmission

[TO WRITE: State money transmitter licensing. FinCEN MSB registration. Partner with licensed transmitter vs. own licenses. State-by-state rollout.]

9.3 Healthcare-Specific Regulation

[TO WRITE: HIPAA compliance architecture. Anti-Kickback Statute analysis. Stark Law considerations. State insurance regulations. ERISA implications.]

9.4 Tax Treatment

[TO WRITE: Tax treatment of zCARE and ZPH for patients, providers, and employers. IRS virtual currency guidance. Staking rewards taxation.]

10. Go-to-Market Strategy

10.1 Phase 1: Token Launch & Community Building (IN PROGRESS)

Phase 1 covers ZPH deployment, ecosystem infrastructure, brand development, community building, and exchange listing. The token launched on March 17, 2026. The following summarizes current status:

10.1.1 Deployment & Listings

- ✓ ZPH SPL token deployed via Smithii.io on Solana mainnet (March 17, 2026)
- ✓ Token metadata hosted on IPFS via Pinata — immutable, Phantom-compatible
- ✓ All three authorities (Mint, Freeze, Update) permanently revoked
- ✓ Raydium ZPH/SOL liquidity pool created and funded
- ✓ Jupiter aggregator listing active
- ✓ DexScreener tracking live
- ✓ CoinGecko listing form submitted (with accurate holder concentration disclosure)
- ✓ Jupiter VRFD (verified token list) submission completed
- ○ CoinGecko listing activation — pending TVL threshold (~\$5,000 required; ~\$1,200 at last check)

10.1.2 Website & Brand

- ✓ Project website live at zypherthecoin.github.io/Zypher
- ✓ Full How-to-Buy guide published (6-step walkthrough: wallet setup → SOL purchase → Jupiter swap)
- ✓ Brand system unified — canonical color #7FF5F0 (cyan), Syne Bold 800 typography
- ✓ Brand asset suite complete: wordmark, lockup, favicon, OG share card, Twitter header, GitHub social preview
- ✓ GitHub repository live at github.com/zypherthecoin/Zypher

10.1.3 Community & Social

- ✓ Twitter/X account active at @zypherthecoin
- ✓ 30-day Twitter content calendar built (60 tweets across 4 themes: healthcare infrastructure thesis, Solana/DEX trading, zCARE Phase 2 teasers, community building)
- ✓ Buffer selected as scheduling platform for content calendar execution
- ✓ Telegram announcement channel established (@ZypherZPH)
- ✓ Telegram community group established (@ZypherCommunity), linked to announcement channel
- ○ Telegram buy-alert bot integration (@OfficialSolanaBuyBot or @MaestroBuyBot) — in progress

10.1.4 Merchandise & Brand Extensions

- ✓ Hat designs completed with Printful embroidery specs (badge + ZPH ticker placements)
- ○ Printful merchandise store — in progress

10.1.5 Liquidity & TVL Growth

The current priority for Phase 1 completion is growing TVL in the Raydium ZPH/SOL pool to the ~\$5,000 threshold required for CoinGecko listing activation. TVL as of last check was approximately \$1,200, requiring approximately \$3,800 in additional liquidity. Volume activity on the pool also remains an active focus, as zero-transaction periods negatively impact optics for exchange reviewers.

Key strategies in execution:

- LP recruitment outreach (mission-forward and direct-ask variants drafted for warm and new contacts)
- Social content calendar driving awareness and buy activity via @zypherthecoin
- Lump-sum liquidity injections prioritized over incremental contributions for maximum listing impact

10.2 Phase 2: Proof of Concept

[TO WRITE: Target DFW metropolitan area. Partner with 3–5 independent primary care practices and 1–2 FQHCs. Deploy zCARE custom Anchor program. Patient cohort: self-pay and high-deductible plan patients. 6–12 month pilot. Success metrics: provider adoption rate, patient satisfaction, transaction volume, redemption patterns.]

10.3 Phase 3: Regional Expansion

Phase 3 scales the validated pilot beyond its initial footprint. The protocol expands across multiple Texas metropolitan areas, broadens beyond primary care to add specialty providers, and introduces employer-sponsored zCARE programs (§5.2) as a patient-funding channel. ZPH governance and staking go live, transitioning the community from passive holders to active participants. Target metrics for the phase are 50–100 participating providers and 5,000–10,000 active patients.

10.4 Phase 4: National Scale

Phase 4 takes Zypher multi-state. The protocol pursues health-system partnerships to onboard providers at scale, begins payer integration to coordinate zCARE with existing coverage, and launches the full provider intelligence layer (§7) as a standalone value proposition. Target metrics are 1,000+ providers and 100,000+ patients, with the operating entity transitioning toward broader DAO governance over the phase.

10.5 Phase 5: Ecosystem Maturity

Phase 5 extends Zypher from a domestic payment rail into a mature healthcare-finance ecosystem. Initiatives include cross-border care payments for medical travel and remittance-funded care, specialty token pools tailored to defined care categories, provider-to-provider settlement for referrals and shared-care arrangements, and a research data marketplace built on the intelligence layer. Governance completes its transition to the DAO, with the founding entity stepping back to a contributor role.

11. Team, Governance & Legal Structure

11.1 Founding Team

[TO WRITE: Team bios, relevant experience in healthcare data, credentialing, provider networks, and blockchain/fintech. Advisory board targets.]

11.2 Legal Entity Structure

[TO WRITE: Foundation structure. Operating company. DAO transition roadmap. IP ownership and licensing.]

11.3 Progressive Decentralization

[TO WRITE: Governance transition timeline. Solana Realms-based DAO implementation. Voting mechanisms, quorum requirements, proposal lifecycle.]

12. Financial Projections & Funding

12.1 Revenue Model

[TO WRITE: Revenue streams and 5-year projection. Reference Zypher_Tokenomics_Model_v02.xlsx.]

12.2 Funding Requirements

[TO WRITE: Phase 1 costs (deployment + Raydium liquidity: ~\$300–\$500 completed). Phase 2 Anchor program engineering investment. Seed round, Series A timeline. Use of funds breakdown.]

12.3 Token Generation Event

The Token Generation Event for ZPH was completed on March 17, 2026. ZPH was deployed via Smithii.io on Solana mainnet. Initial liquidity was seeded in a Raydium ZPH/SOL pool. The token is live, tradeable, and listed on Jupiter aggregator. Future CEX listing strategy will follow CoinGecko and Jupiter verified listing milestones.

13. Risk Factors & Mitigations

[TO WRITE: Risk matrix: (a) Regulatory. (b) zCARE stability/depeg. (c) Adoption. (d) Technical (Solana outages, smart contract vulnerability). (e) Competitive. (f) Macro (crypto market). (g) Concentration risk — majority of supply in treasury wallet. For each: likelihood, impact, mitigation.]

14. Conclusion

Zypher addresses healthcare's payment crisis at the protocol level rather than the product level. Its dual-token architecture separates the stability that providers require (zCARE) from the ecosystem dynamics that drive network growth (ZPH), so that neither compromises the other. Grounded in real healthcare data and verified provider identities, and built on Solana's high-throughput, low-cost settlement, Zypher realigns the incentives between patients who cannot fully pay and providers who cannot afford to absorb the loss.

For underinsured patients, it offers a funded, dignified way to compensate their providers. For providers, a guaranteed, low-cost settlement channel and a window into the data and credentialing infrastructure they already need. For the broader community, a transparent, governable public good. We invite patients, providers, employers, grant-makers, and builders to participate in establishing a payment layer worthy of the care it settles.

Appendices

Appendix A: Glossary of Terms

Healthcare Terms

Term	Definition
NPI	National Provider Identifier — a unique 10-digit identifier issued to U.S. healthcare providers.
NPES	National Plan and Provider Enumeration System — the CMS system that issues and maintains NPIs.
CPT	Current Procedural Terminology — standardized codes describing medical procedures and services.
HCPCS	Healthcare Common Procedure Coding System — codes for products, supplies, and services not covered by CPT.
RVU	Relative Value Unit — the measure used in Medicare's fee schedule to value clinical services.
CMS	Centers for Medicare & Medicaid Services.
FQHC	Federally Qualified Health Center — a community-based provider receiving HRSA funding to serve underserved populations.
HPSA	Health Professional Shortage Area — a federal designation for regions with provider shortages.
MUA	Medically Underserved Area — a federal designation for geographic areas with inadequate access to primary care.
HRSA	Health Resources and Services Administration.
EOB	Explanation of Benefits — a statement describing what a health plan covered for a service.

Solana / Crypto Terms

Term	Definition
SPL Token	Solana Program Library token — Solana's standard fungible-token format.
PDA	Program Derived Address — a deterministic, program-owned Solana account address.
Anchor	A Rust framework for building Solana programs.
Raydium	A Solana automated market maker and liquidity provider.
Jupiter	A Solana liquidity aggregator that routes trades for best execution.
Pyth	A high-frequency on-chain oracle network providing real-time price feeds.
Switchboard	A decentralized Solana oracle used here as a fallback price source.
Stablecoin	A token designed to maintain a stable value, typically against the U.S. dollar.

Term	Definition
DAO	Decentralized Autonomous Organization — an on-chain, token-holder-governed organization.
TVL	Total Value Locked — the total value of assets deposited in a DeFi protocol or liquidity pool.
LP	Liquidity Provider — a participant who deposits assets into a pool to facilitate trading.
IPFS	InterPlanetary File System — a decentralized content-addressed storage network used for token metadata hosting.

Appendix B: Detailed Tokenomics Model

[TO WRITE: Reference Zypher_Tokenomics_Model_v02.xlsx. Supply schedules, fee projections, burn rates, scenario analysis.]

Appendix C: Solana Deployment Record

This appendix documents the complete deployment history for ZPH on Solana mainnet, including an initial deployment that required remediation and the subsequent clean redeployment. This record is preserved for transparency and as a reference for teams building on Solana.

C.1 Initial Deployment — Issues Encountered

The initial ZPH deployment encountered two critical errors in sequence that made the token unrecoverable without a full redeployment.

Issue	Description	Impact
Imgur-Hosted Logo	Original token metadata pointed to an Imgur-hosted image URL. Phantom Wallet does not recognize CDN or third-party image hosts as trusted sources for token logos.	Token flagged as spam in Phantom — effectively unusable for most users.
Premature Authority Revocation	Before the logo issue was identified and corrected, the Update Authority had already been revoked. On Solana, this makes on-chain metadata permanently immutable.	The incorrect Imgur URL was locked into the token permanently with no on-chain repair path.
No Recovery Path	Immutable metadata + active spam flag left no viable repair mechanism.	Full redeployment with a new mint address was the only path forward.

C.2 Clean Redeployment — Corrected Workflow

Step	Action	Result
1	New token created via Smithii.io	Fresh SPL token minted — new mint address
2	Logo resized to 512×512px, transparent background, circular mask applied	Specification-compliant image asset
3	Logo + metadata JSON uploaded to Pinata (IPFS)	Immutable, wallet-compatible hosting. CID: bafkreihn5krmef7sbxnsvnav7clvp6on3h5edarl7nudbvvyjlgxz4zx6q
4	On-chain metadata updated to IPFS URI and fully verified	Phantom-compatible — no spam flag. Verified before any revocation.
5	Mint Authority revoked	Fixed supply — no additional ZPH can ever be minted
6	Freeze Authority revoked	Token accounts cannot be frozen by any party
7	Update Authority revoked	Metadata permanently immutable — correct IPFS URI locked in
8	Raydium ZPH/SOL liquidity pool created	Pool ID: FKsYM7mBRR3oFPnJZYNYzSu9MjJLC4kYxatzYtPq8Aba
9	Jupiter aggregator listing activated	ZPH discoverable at jup.ag
10	DexScreener tracking confirmed	Price and volume publicly visible
11	Project website launched	https://zypherthecoin.github.io/Zypher/
12	CoinGecko listing submitted	Under review — pending TVL threshold
13	Jupiter VRFD submitted	Under review
14	Telegram channels established	t.me/ZypherZPH (announcements) + t.me/ZypherCommunity (community)

C.3 Current Token Details

Parameter	Value
Mint Address	8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t
Pinata IPFS CID	bafkreihn5krmef7sbxnsvnav7clvp6on3h5edarl7nudbvvyjlgxz4zx6q
Raydium Pool ID	FKsYM7mBRR3oFPnJZYNYzSu9MjJLC4kYxatzYtPq8Aba
Jupiter	https://jup.ag/tokens/8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t

Parameter	Value
DexScreener	https://dexscreener.com/solana/8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t
Website	https://zypherthecoin.github.io/Zypher/
GitHub	https://github.com/zypherthecoin/Zypher
Twitter	@zypherthecoin
Telegram Announcements	t.me/ZypherZPH
Telegram Community	t.me/ZypherCommunity
Solana Explorer	https://explorer.solana.com/address/8sG1MjujqWKNZKkkGQQdekuMTLJvFdP3hoohyEiUr82t
Mint Authority	Revoked
Freeze Authority	Revoked
Update Authority	Revoked

C.4 Key Lessons Learned

- **Use IPFS:** Token logos must be hosted on IPFS (e.g., Pinata). CDN/image host URLs (Imgur, etc.) trigger Phantom spam flags. IPFS is the only accepted standard for Solana token metadata images.
- **Verify Before Revoking:** On-chain metadata must be fully verified and confirmed correct before the Update Authority is revoked. Once revoked, metadata is permanently immutable — there is no patch path.
- **Sequencing Is Non-Negotiable:** Errors discovered after authority revocation cannot be fixed. A new token deployment is the only remedy, resetting all listings, liquidity positions, and community awareness. Prevention is the only mitigation.
- **Listing Prerequisites:** CoinGecko listing submissions effectively require: a live project website, meaningful liquidity and trading volume, a whitepaper, and accurate holder/supply data. Sequence: deploy → verify → website → liquidity → volume → submission.

Appendix D: Regulatory Jurisdiction Analysis

[TO WRITE: State-by-state regulatory landscape for money transmission and digital asset classification.]

Appendix E: Competitive Landscape

[TO WRITE: Healthcare payment innovations, healthcare-adjacent crypto projects (MedToken, Solve.Care, Patientory), and traditional fintech competitors.]

Appendix F: Brand Identity Standards

The following design standards govern all Zypher visual communications:

Element	Specification
Primary Color	#7FF5F0 (Cyan) — canonical brand color
Typography	Syne Bold 800 — primary headlines and wordmarks
Logo Format	512×512px PNG, transparent background, circular mask
Twitter / X Handle	@zypherthecoin
GitHub Repository	github.com/zypherthecoin/Zypher
Website	zypherthecoin.github.io/Zypher
OG Share Card	1200×630px, cyan brand treatment
Twitter Header	1500×500px, cyan brand treatment
GitHub Social Preview	1280×640px, cyan brand treatment